

Test 4 — Question Paper

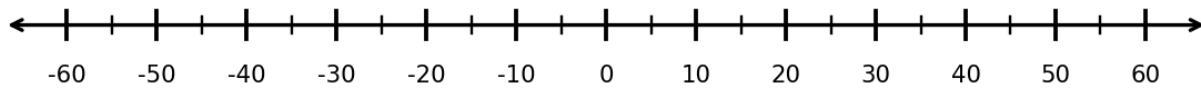
38 items. Suggested time: 95 minutes (about 2 minutes 30 seconds per item).

A four-function calculator may be used **only** on items marked *calculator*.

The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.

Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. Plot the value of $-35 + 30$ on the number line.



Mark your answer on the diagram above.

Show your work

calculator · graphing (grid)

2. What is the value of $|-20 \frac{3}{10}|$?

- A. -0.05
- B. -20.3
- C. 40.6
- D. 20.3

Show your work

calculator · multiple choice

3. A triangle has an area of 33 square feet and a height of 11 feet.

What is the length of its base, in feet?

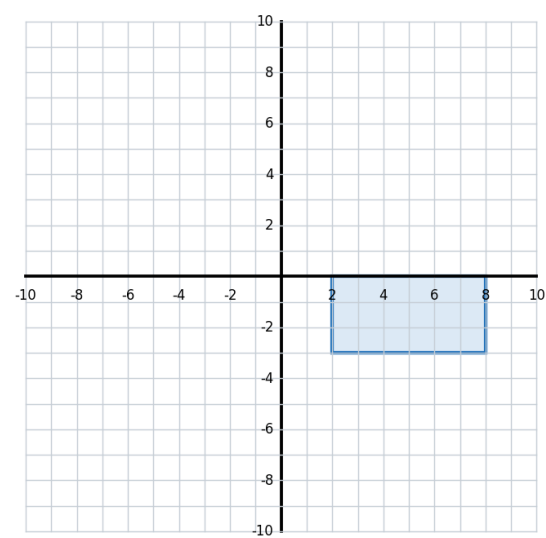
- A. 8
- B. 3
- C. 66
- D. 6

Show your work

multiple choice

4. A rectangle has vertices at (2, -3), (8, -3), (8, 0) and (2, 0).

What is the area, in square units, of the rectangle?



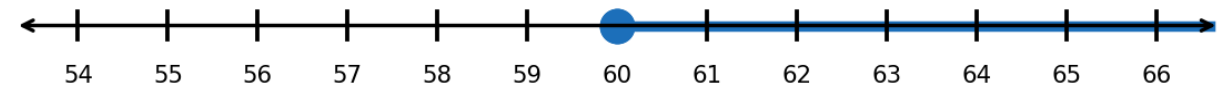
- A. 18
- B. 9
- C. 21
- D. 36

Show your work

multiple choice

5. The number line shows the solution to an inequality.

Which inequality does the graph represent?



- A. $x \geq 60$
- B. $x < 60$
- C. $x \leq 60$
- D. $x > 60$

Show your work

multiple choice

6. Nico wrote the expression $x - 8$.

Which written description matches the expression?

- A. a number x divided by 8
- B. 8 less than a number x
- C. the product of a number x and 8
- D. a number x decreased by 8

Show your work

multiple choice

7. Elena recorded these four numbers in her math journal: -3.5 , $-3/4$, $1/5$, 3 .

Which number is the least?

- A. $1/5$
- B. $-3/4$
- C. -3.5
- D. 3

Show your work

calculator · multiple choice

8. A box plot of a data set has the five-number summary shown below the table.

Match each part of the summary with its value.

	6	11	15	18
Minimum	■	■	■	■
Median	■	■	■	■
Upper quartile	■	■	■	■

Show your work

matching table

9. Use the drop-down menus to complete the solutions.

If $x + 3.54 = 8.77$, then $x = \text{[[1]]}$.

If $x - 3.54 = 5.23$, then $x = \text{[[2]]}$.

Blank (1) choices: 12.31 | 3.54 | 5.23

Blank (2) choices: 1.69 | 5.23 | 8.77

Show your work

editing task choice

10. Match each right rectangular prism with its volume, in cubic inches.

	24	45	60	66
2 by 3 by 4 inches	■	■	■	■
2 by 5 by 6 inches	■	■	■	■
3 by 3 by 5 inches	■	■	■	■

Show your work

matching table

11. Use the drop-down menus to give each solution.

If $x + 11 = 22$, then $x = \text{[[1]]}$.

If $x - 27 = 11$, then $x = \text{[[2]]}$.

Blank (1) choices: 11 | 22 | 33

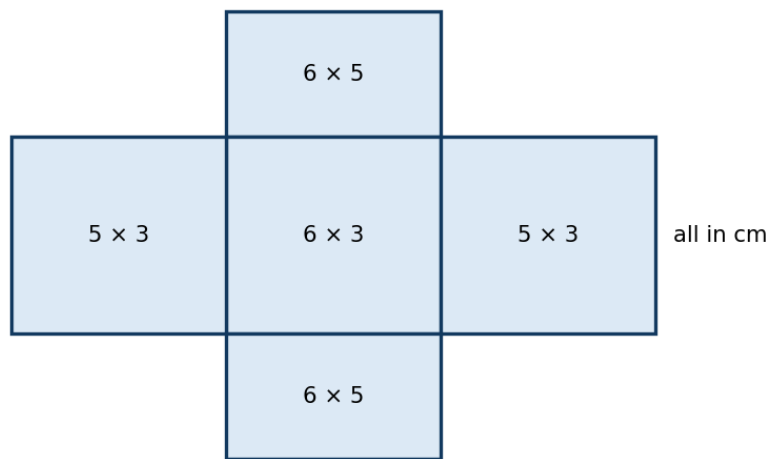
Blank (2) choices: -16 | 11 | 27 | 38

Show your work

editing task choice

12. Use the drop-down menus to find the surface area of the prism from its net.

The two 6 by 5 faces have a total area of square centimeters, and the two 6 by 3 faces have a total area of square centimeters.



Blank (1) choices: 11 | 30 | 60 | 120

Blank (2) choices: 9 | 18 | 36 | 72

Show your work

editing task choice

13. The table shows equivalent ratios of buses to vans.

Use the drop-down menus to complete the table: the first ? is and the second ? is .

Buses	Vans
3	5
?	10
9	?

Blank (1) choices: 3 | 5 | 6 | 10

Blank (2) choices: 5 | 8 | 9 | 15

Show your work

editing task choice

14. A parking lot has 24 cars and 15 trucks.

Use the drop-down menus to write the ratio of cars to trucks in simplest form.

[[1]] : [[2]]

Blank (1) choices: 5 | 8 | 9 | 24

Blank (2) choices: 5 | 6 | 8 | 15

Show your work

editing task choice

15. Use the drop-down menus to complete the statements about 240 students.

20% of 240 is [[1]] students, and 50% of 240 is [[2]] students.

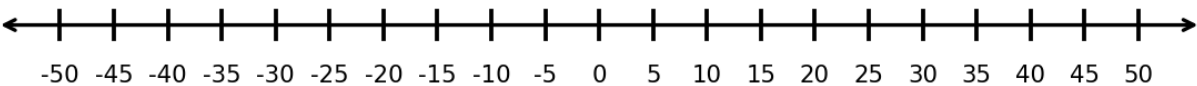
Blank (1) choices: 20 | 48 | 96 | 220

Blank (2) choices: 60 | 120 | 240 | 480

Show your work

editing task choice

16. Diego and a friend swim to depths with the same absolute value. One is 35 feet below sea level. Plot both depths on the number line.



Mark your answer on the diagram above.

Show your work

calculator · graphing (grid)

17. Match each number with the equivalent percent.

	25%	50%	70%	77%
1/4	☐	☐	☐	☐
0.5	☐	☐	☐	☐
7/10	☐	☐	☐	☐

Show your work

calculator · matching table

18. Point A is located at (-3, -5). Point A is reflected across the y-axis.

What are the coordinates of the reflected point?

- A. (3, 5)
- B. (3, -5)
- C. (-3, -5)
- D. (-5, -3)

Show your work

multiple choice

19. A diver off Crandon Beach descends 25 feet below the surface, then rises 11 feet. Sea level is represented by 0.

Which integer represents the diver's position now?

- A. -14
- B. 14
- C. 36
- D. -36

Show your work

calculator · multiple choice

20. Use the drop-down menus to rewrite $25 + 35$ using the greatest common factor.

$25 + 35 = \text{[[1]]} \times (\text{[[2]]} + \text{[[3]]})$

Blank (1) choices: 2 | 5 | 10

Blank (2) choices: 5 | 6 | 7 | 25

Blank (3) choices: 5 | 7 | 8 | 35

Show your work

calculator · editing task choice

21. Match each power with its value.

	4	15	25	125
5^2	☐	☐	☐	☐
5^3	☐	☐	☐	☐
2^2	☐	☐	☐	☐

Show your work

calculator · matching table

22. What is the value of $(2x + 2) \div 4$ when $x = 7$?

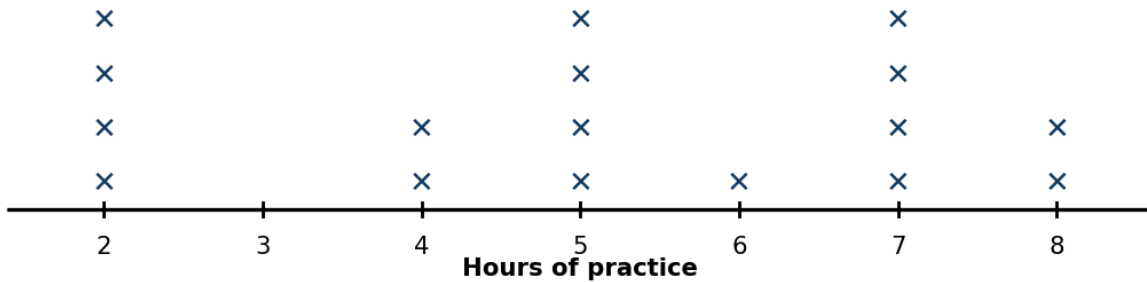
- A. 16
- B. 4
- C. 5
- D. 14

Show your work

multiple choice

23. The line plot shows hours of practice.

How many students are represented on the line plot? Enter your answer.



Answer: _____

Show your work

equation editor

24. Four sensors on a reef near Key Biscayne report these elevations, in feet relative to sea level:

-63, -47, -25, -12

Which elevation is farthest from sea level?

- A. -12
- B. -63
- C. -25
- D. -47

Show your work

calculator · multiple choice

25. Sofia has \$38.75 and spends \$5.80 on a bus pass. The rest is shared equally among 5 people.

After buying the pass, Sofia has left, and each person receives .

Blank (1) choices: \$32.95 | \$44.55 | \$9.75

Blank (2) choices: \$164.75 | \$6.59 | \$7.75

Show your work

calculator · editing task choice

26. Use the drop-down menus to complete the two statements.

$27 \times 0.25 = \text{[[1]]}$ and $27 \div 0.25 = \text{[[2]]}$

Blank (1) choices: 108 | 26.75 | 27.25 | 6.75

Blank (2) choices: 108 | 26.75 | 27.25 | 6.75

Show your work

calculator · editing task choice

27. A data set is shown.

2, 3, 4, 10, 12, 13, 14, 16, 16, 18, 23

How many values belong in the interval 19-24 of a histogram? Enter your answer.

Answer: _____

Show your work

equation editor

28. For each value of x, decide whether $x \geq 20$ is true or false.

	True	False
15	☐	☐
18	☐	☐
20	☐	☐
29	☐	☐

Show your work

matching table

29. Use the drop-down menus to complete the equations.

$(-7)(-8) = \text{[[1]]}$ and $(56) \div (-8) = \text{[[2]]}$

Blank (1) choices: -56 | -15 | 1 | 56

Blank (2) choices: -8 | -7 | 7 | 56

Show your work

calculator · editing task choice

30. Match each expression with its value.

	10/27	15/8	23/18	8/15
$5/6 \times 4/9$	☐	☐	☐	☐
$5/6 \div 4/9$	☐	☐	☐	☐
$4/9 \div 5/6$	☐	☐	☐	☐

Show your work

calculator · matching table

31. Match each description with its value.

	12	30	36	6
GCF of 12 and 18	☐	☐	☐	☐
LCM of 12 and 18	☐	☐	☐	☐
GCF of 12 and 36	☐	☐	☐	☐

Show your work

calculator · matching table

32. Match each measurement with its equivalent value. Use the Grade 6 reference sheet.

	12	32	50	500
2 pounds, in ounces	☐	☐	☐	☐
3 gallons, in quarts	☐	☐	☐	☐
5 meters, in centimeters	☐	☐	☐	☐

Show your work

matching table

33. A data set is shown.

8, 9, 12, 17, 24

After 40 is added to the data set, match each measure with its new value.

	14 1/2	18 1/3	32	35
Mean	☐	☐	☐	☐
Median	☐	☐	☐	☐
Range	☐	☐	☐	☐

Show your work

matching table

34. Select **all** the questions below that would generate numerical data with variability.

Circle every correct choice:

- ☐ How many text messages did each student send yesterday?
- ☐ What are the heights, in inches, of the students in the band?
- ☐ How many books did I read last month?
- ☐ How old is my teacher?
- ☐ How many minutes long is my bus ride to school?

Show your work

selectable hot text

35. Select **all** the prime factors of 50.

Circle every correct choice:

- ☐ 1
- ☐ 2
- ☐ 4
- ☐ 5
- ☐ 50

Show your work

calculator · selectable hot text

36. A data set is shown.

11, 15, 17, 20, 22, 23

Match each measure with its value.

	12	18	18 1/2	20 1/2
Mean	☐	☐	☐	☐
Median	☐	☐	☐	☐
Range	☐	☐	☐	☐

Show your work

matching table

37. Match each rate with its unit rate (the amount for exactly 1 unit).

	4	9	12	14
16 miles in 4 hours	☐	☐	☐	☐
24 dollars for 2 pounds	☐	☐	☐	☐
54 pages in 6 minutes	☐	☐	☐	☐

Show your work

matching table

38. Use the drop-down menus to give each solution.

If $4x = 36$, then $x =$ [[1]].

If $x \div 2 = 9$, then $x =$ [[2]].

Blank (1) choices: 4 | 9 | 36 | 144

Blank (2) choices: 2 | 4 | 9 | 18

Show your work

editing task choice

Grade 6 Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches	1 cup = 8 fluid ounces
1 yard = 3 feet	1 pint = 2 cups
1 mile = 5,280 feet	1 quart = 2 pints
1 mile = 1,760 yards	1 gallon = 4 quarts
1 pound = 16 ounces	1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters	1 liter = 1,000 milliliters
1 meter = 1,000 millimeters	1 gram = 1,000 milligrams
1 kilometer = 1,000 meters	1 kilogram = 1,000 grams

Time Conversions

1 minute = 60 seconds	1 week = 7 days
1 hour = 60 minutes	1 year = 365 days
1 day = 24 hours	1 year = 52 weeks

Formula

Rectangular prism	$V = lwh$ or $V = Bh$
l = length, w = width, h = height	B = area of base, V = volume